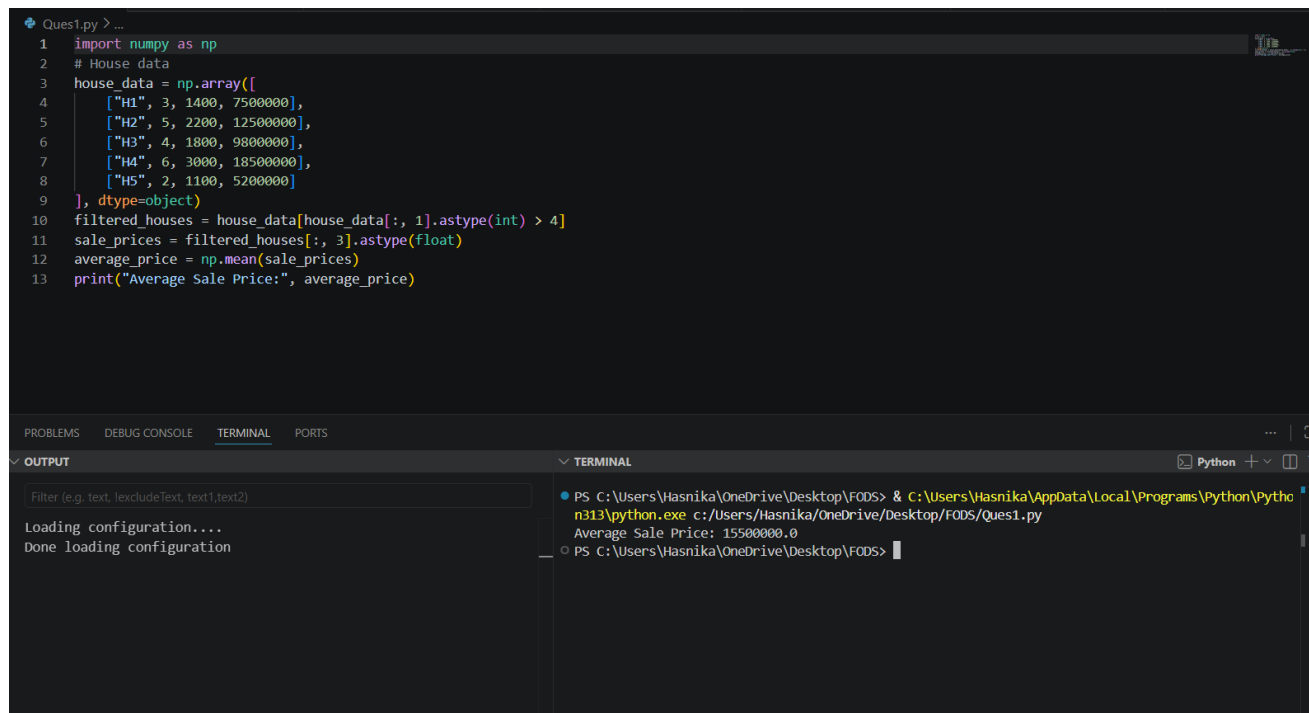


List of Scenario to develop Python code and Generate Output

Question1:

Using NumPy arrays and operations, how would you find the average sale price of houses with more than four bedrooms in the neighborhood?



```
1 import numpy as np
2 # House data
3 house_data = np.array([
4     ["H1", 3, 1400, 7500000],
5     ["H2", 5, 2200, 12500000],
6     ["H3", 4, 1800, 9800000],
7     ["H4", 6, 3000, 18500000],
8     ["H5", 2, 1100, 5200000]
9 ], dtype=object)
10 filtered_houses = house_data[house_data[:, 1].astype(int) > 4]
11 sale_prices = filtered_houses[:, 3].astype(float)
12 average_price = np.mean(sale_prices)
13 print("Average Sale Price:", average_price)
```

OUTPUT

Loading configuration....
Done loading configuration

TERMINAL

```
PS C:\Users\Hasnika\OneDrive\Desktop\FODS> & C:\Users\Hasnika\AppData\Local\Programs\Python\Python313\python.exe c:/Users/Hasnika/OneDrive/Desktop/FODS/Ques1.py
Average Sale Price: 15500000.0
PS C:\Users\Hasnika\OneDrive\Desktop\FODS>
```

Question2:

Develop a Logistic Regression model using the given features.

Predict whether a newly received email is spam or not.

Email_ID Num_Links Num_Caps_Words Email_Length Spam or not

```
Ques2.py > ...
17 model = LogisticRegression()
18 model.fit(X, y)
19
20 # New email to predict
21 new_email = np.array([[5, 12, 280]])
22
23 # Predict
24 prediction = model.predict(new_email)
25
26 # Display result
27 if prediction[0] == 1:
28     print("Spam Email")
29 else:
30     print("Not Spam")
```

PROBLEMS DEBUG CONSOLE TERMINAL PORTS

OUTPUT

Filter (e.g. text, excludeText, text1, text2)

loading configuration...
Done loading configuration

TERMINAL

Python +

- PS C:\Users\Hasnika\OneDrive\Desktop\FODS> & C:\Users\Hasnika\AppData\Local\Programs\Python\Python313\python.exe c:/Users/Hasnika/OneDrive/Desktop/FODS/Ques2.py
Spam Email
- PS C:\Users\Hasnika\OneDrive\Desktop\FODS>

Question4:

Write a Python program that loads the dataset containing Age, Kms_Driven, Engine_CC, and Resale_Price, trains a Linear Regression model using scikit-learn, accepts user input for a new car, and predicts the estimated resale price.

```
Ques4.py > ...
1 from sklearn.linear_model import LinearRegression
2 import numpy as np
3 X = np.array([
4     [2, 25000, 1200],
5     [5, 60000, 1500],
6     [3, 40000, 1300],
7     [7, 90000, 1800],
8     [1, 15000, 1000]
9 ])
10 y = np.array([650000, 420000, 520000, 300000, 720000])
11 model = LinearRegression()
12 model.fit(X, y)
13 new_car = np.array([[4, 50000, 1400]])
14 prediction = model.predict(new_car)
15 print("Predicted Resale Price = ₹", prediction[0])
```

PROBLEMS DEBUG CONSOLE TERMINAL PORTS

OUTPUT

Filter (e.g. text, excludeText, text1, text2)

Loading configuration...
Done loading configuration

TERMINAL

Python +

- PS C:\Users\Hasnika\OneDrive\Desktop\FODS> & C:\Users\Hasnika\AppData\Local\Programs\Python\Python313\python.exe c:/Users/Hasnika/OneDrive/Desktop/FODS/Ques4.py
Predicted Resale Price = ₹ 495263.15789473685
- PS C:\Users\Hasnika\OneDrive\Desktop\FODS>

Question5:

Using CART, build a Decision Tree Classifier for loan approval. Predict the class for a new applicant and display the decision rules.

 Python ...

```
PS C:\Users\Hasnika\OneDrive\Desktop\FODS> & C:\Users\Hasnika\AppData\Local\Programs\Python\Python313\python.exe c:/Users/Hasnika/OneDrive/Desktop/FODS/Ques5.py
Loan Approved
```

```
|--- Credit_Score <= 680.00
|   |--- class: 0
|--- Credit_Score > 680.00
|   |--- class: 1
```